

A REAL TIME PROJECT / FIELD BASED PROJECT ON

TRUCK PLATOONING SYSTEM

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

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CERTIFICATE

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"TRUCK PLATOONING SYSTEM"

under the supervision and guidance of Mr. T. Mohan Das in fulfilment of the requirements of the fourth semester, BACHELOR OF TECHNOLOGY, ELECTRONICS & COMMUNICATION ENGINEERING in JNTUH UNIVERSITY COLLEGE OF ENGINEERING SULTANPUR during the academic year 2025–2026.

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

TRUCK PLATOONING SYSTEM

The Truck Platooning System is a semi-autonomous logistics solution designed to optimise large-scale transportation by enabling multiple vehicles to move in a coordinated convoy under the control of a single lead vehicle. In this system, the lead truck is manually driven, while the follower trucks operate autonomously by continuously tracking the leader's motion parameters.

Each follower vehicle is equipped with distance-measuring sensors (HC-SR04 Ultrasonic Sensor), directional sensing units (IR Sensors), and a wireless communication module (HC-05 Bluetooth) to receive real-time data related to the speed, direction, and movement of the lead vehicle. The onboard microcontroller (Arduino Uno) processes real-time sensor feedback to maintain a predefined safe inter-vehicle distance, synchronise velocity, and ensure directional alignment.

Motor control algorithms implemented through the L298N motor driver dynamically adjust speed and steering to achieve smooth and stable platoon movement while preventing collisions. Additionally, the system incorporates safety mechanisms such as obstacle detection and emergency stopping to improve operational reliability.

By reducing the requirement for individual drivers in every vehicle, the Truck Platooning System enhances transportation efficiency, lowers manpower costs, and improves fuel utilization. This system is well suited for applications in logistics transportation, mining operations, industrial material handling, and long-distance freight convoy systems.

Key Components:

- Arduino Uno – Main microcontroller for processing and control
- L298N Motor Driver Module – Motor speed and direction control
- HC-SR04 Ultrasonic Sensor – Distance measurement for safe following
- IR Sensors (Left & Right) – Directional tracking and lane guidance
- HC-05 Bluetooth Module – Wireless communication between vehicles
- DC Motors (2x) with Robot Chassis – Locomotion system
- 7.4V / 9V / 12V Battery – Power supply

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CHAPTER 1: INTRODUCTION

1.1 Introduction to Truck Platooning

The transportation and logistics industry is one of the largest consumers of fuel and human resources worldwide. With growing demand for efficient freight movement across long distances, there is a pressing need for technologies that reduce operational costs while improving safety. Truck platooning is an emerging autonomous driving concept in which a group of trucks travel together in a convoy, maintaining close and consistent spacing, governed by wireless communication and automated control systems.

This project presents the design and implementation of a Truck Platooning System using Arduino Uno microcontrollers, Ultrasonic distance sensors (HC-SR04), IR sensors, a Bluetooth communication module (HC-05), and an L298N motor driver. The system is developed on a prototype robot chassis to simulate real-world truck platooning behavior.

In this system, the leader truck is manually driven or controlled via Bluetooth commands from a mobile phone, while the follower truck autonomously replicates the leader's motion by continuously measuring the inter-vehicle distance and adjusting its speed and direction accordingly. This eliminates the need for a dedicated driver in each following vehicle, making the convoy safer, more fuel-efficient, and cost-effective.

Key Concept

Truck Platooning reduces aerodynamic drag on follower vehicles by maintaining close convoy spacing, resulting in fuel savings of 10–25% for follower trucks. The leader truck saves approximately 5% fuel due to reduced air resistance from vehicles behind it.

1.2 Aim and Objectives

The main aim of this project is to design and implement a Truck Platooning System prototype using Arduino Uno that allows multiple vehicles to move in a coordinated convoy with minimal human intervention.

The objectives of this project include:

- To design a semi-autonomous follower vehicle that tracks a leader vehicle in real time.
- To implement distance control using the HC-SR04 Ultrasonic Sensor for safe inter-vehicle spacing.
- To use IR sensors for directional guidance and obstacle avoidance on the follower vehicle.
- To establish Bluetooth-based wireless communication between the leader and follower vehicles.
- To control DC motor speed and direction using the L298N motor driver module with Arduino PWM signals.
- To develop a simple, cost-effective, and scalable prototype of a truck platooning system.
- To demonstrate the principles of autonomous vehicle control and embedded systems integration.

1.3 Methodology

The development of the Truck Platooning System is carried out in a systematic and phased manner involving hardware selection, circuit design, programming, integration, and testing.

Phase 1 – Component Selection and Procurement:

The required components — Arduino Uno, L298N motor driver, HC-SR04 ultrasonic sensor, IR sensors, HC-05 Bluetooth module, DC motors, robot chassis, and power supply — are carefully selected based on compatibility, availability, and cost.

Phase 2 – Circuit Design and Hardware Assembly:

The circuit is designed by connecting all sensors and actuators to the Arduino Uno. The HC-05 Bluetooth module is interfaced via UART serial communication. The ultrasonic sensor is connected to digital I/O pins for trigger and echo signals. IR sensors are connected to analog/digital input pins. The L298N motor driver is connected to PWM-capable pins of the Arduino for speed control. The entire circuit is assembled on the robot chassis.

Phase 3 – Software Development:

The Arduino Uno is programmed using the Arduino IDE in C/C++. The program is developed to read sensor data, process Bluetooth commands from the leader vehicle, compute the appropriate motor control signals, and implement the platooning logic including safe-distance maintenance, speed synchronization, and emergency stopping.

Phase 4 – Testing and Calibration:

After integration, the system is tested individually for each subsystem — Bluetooth communication, ultrasonic distance measurement, IR-based direction tracking, and motor control. The system is then tested as a whole in platooning mode. Calibration is performed to optimize following distance thresholds and speed response.

1.4 Significance of the Project

The Truck Platooning System addresses several important real-world challenges in logistics and transportation:

Fuel Efficiency:

By maintaining close convoy spacing, follower vehicles experience significantly reduced aerodynamic drag. Studies have shown that platooning can reduce fuel consumption by up to 15–25% for follower trucks.

Safety Enhancement:

Automated inter-vehicle distance maintenance and obstacle detection reduce the risk of rear-end collisions. The system reacts faster than human drivers, making the convoy inherently safer.

Reduced Manpower Requirements:

With only the lead truck requiring a human driver, operating costs are drastically reduced for long-distance freight transport involving multiple vehicles.

Educational and Research Value:

This project serves as a foundation for understanding autonomous vehicle systems, embedded control, wireless communication, and sensor integration — all critical technologies in the modern automotive and robotics industries.

CHAPTER 2: MATERIALS AND METHODS

2.1 Components Overview

This chapter presents the detailed description of all hardware components used in the Truck Platooning System. The system is built around the Arduino Uno microcontroller as the central processing unit. Communication between vehicles is achieved through the HC-05 Bluetooth module. Distance sensing is handled by the HC-SR04 Ultrasonic Sensor, while directional control is assisted by IR sensors. The L298N motor driver module controls two DC motors mounted on the robot chassis. The system is powered by a rechargeable battery pack.

Complete Components List:

S.No	Component	Quantity	Purpose
1	Arduino Uno (ATmega328P)	2	Main microcontroller for leader and follower vehicles
2	L298N Motor Driver Module	2	Controls DC motor speed and direction via PWM
3	HC-SR04 Ultrasonic Sensor	2	Measures inter-vehicle distance (2cm – 400cm range)
4	IR Sensor Module	4 (2 per vehicle)	Left/right obstacle and direction detection
5	HC-05 Bluetooth Module	2	Wireless serial communication between vehicles
6	DC Motor (TT Gear Motor)	4 (2 per vehicle)	Drives the robot chassis wheels
7	Robot Chassis (4-wheel)	2	Mechanical frame for the vehicle prototype
8	Battery Pack (7.4V / 9V)	2	Main power supply for each vehicle unit
9	Toggle / Slide Switch	2	System power ON/OFF control
10	Jumper Wires (Assorted)	1 set	Electrical interconnections between components
11	Breadboard (optional)	1	Prototyping and temporary connections

2.2 Arduino Uno (ATmega328P)

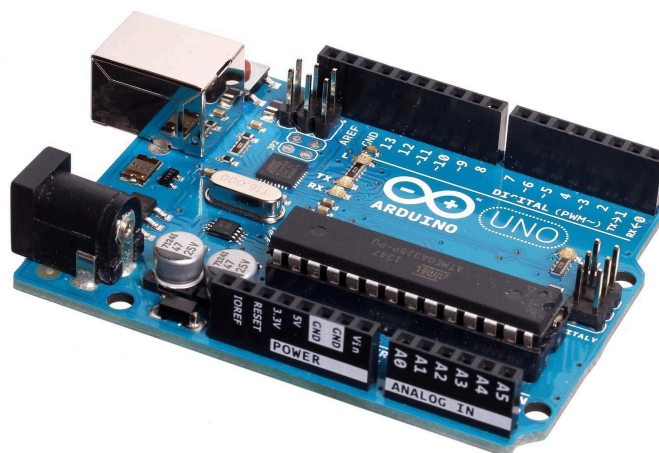
The Arduino Uno is the primary microcontroller used in this project for both the leader and follower vehicles. It is based on the ATmega328P 8-bit microcontroller operating at 16 MHz, featuring 14 digital I/O pins (6 of which support PWM output), 6 analog input pins, 32 KB of flash memory, 2 KB of SRAM, and 1 KB of EEPROM.

Role in the Truck Platooning System:

In the leader vehicle, the Arduino Uno receives movement commands from the user via the HC-05 Bluetooth module and drives the motors accordingly. In the follower vehicle, the Arduino Uno reads data from the HC-SR04 ultrasonic sensor and IR sensors, processes the received distance and direction information, and generates PWM control signals for the L298N motor driver to replicate the leader's movement while maintaining a safe following distance.

Key Specifications:

Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7–12V
Digital I/O Pins	14 (6 PWM capable)
Analog Input Pins	6
DC Current per I/O Pin	20 mA
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
Clock Speed	16 MHz
Communication	UART, SPI, I2C



[Figure 2.1 – Arduino Uno Board]

Note: The Arduino Uno board features a USB-B connector for programming, a barrel jack for power supply, and clearly labelled I/O headers for easy component interfacing.

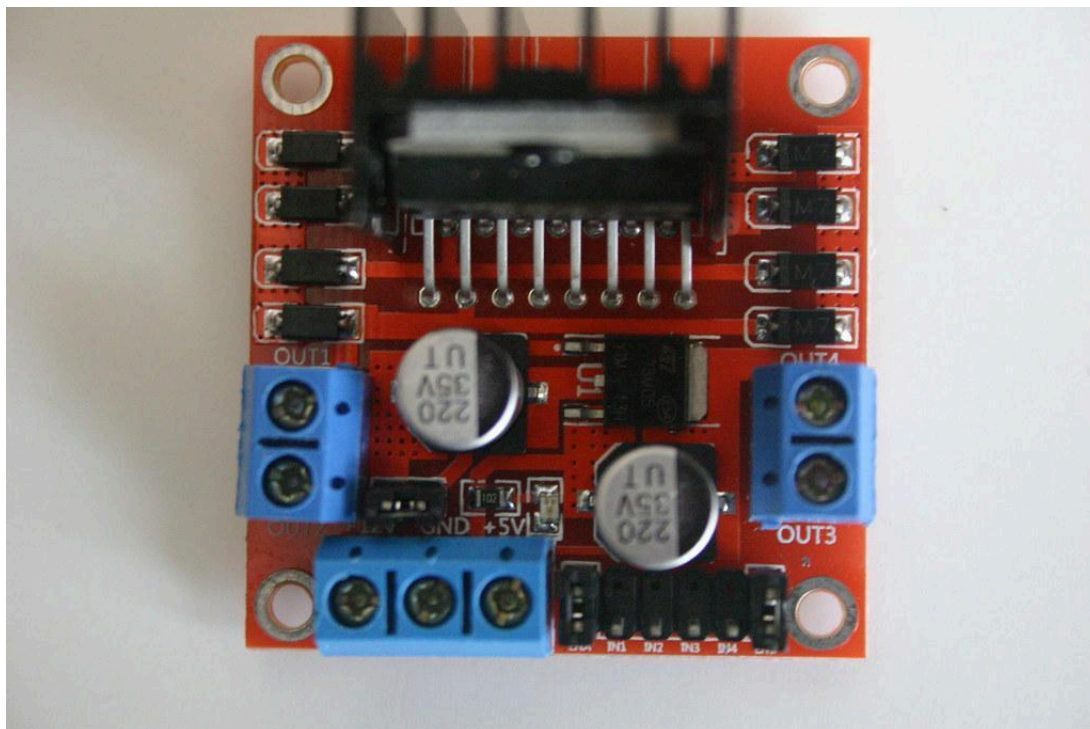
2.3 L298N Motor Driver Module

The L298N is a dual H-bridge motor driver IC capable of driving two DC motors simultaneously with independent speed and direction control. It operates at a supply voltage of 5–35V and can deliver up to 2A per channel (peak 3A), making it suitable for driving small to medium DC motors.

Role in the Truck Platooning System:

The L298N module is connected between the Arduino Uno and the two DC motors on each vehicle. The Arduino sends PWM signals to the enable pins (ENA and ENB) to control motor speed, and HIGH/LOW digital signals to the input pins (IN1, IN2, IN3, IN4) to control motor direction (forward, backward, left turn, right turn, stop).

Driver IC	L298N Dual H-Bridge
Operating Supply Voltage	5V – 35V
Logic Supply Voltage	5V
Max Current per Channel	2A (peak 3A)
Control Logic	TTL Compatible
PWM Frequency	Up to 40 kHz
Protection	Thermal shutdown, over-current
Number of Motors Driven	2 DC motors simultaneously



[Figure 2.2 – L298N Motor Driver Module]

The L298N module includes an onboard 5V regulator, heat sink for thermal management, and screw terminals for motor and power connections.

2.4 HC-SR04 Ultrasonic Sensor

The HC-SR04 is a highly popular, low-cost ultrasonic distance sensor capable of measuring distances from 2 cm to 400 cm with an accuracy of approximately 3 mm. It operates by emitting a 40 kHz ultrasonic pulse and measuring the time taken for the echo to return, which is then converted to distance using the speed of sound formula: $\text{Distance} = (\text{Time} \times 0.0343) / 2$ cm.

Role in the Truck Platooning System:

The HC-SR04 sensor is mounted at the front of the follower truck. It continuously measures the distance between the follower and the leader vehicle. The Arduino compares this measured distance against a predefined threshold (safe following distance). If the follower gets too close, the Arduino slows down or stops the motors. If the distance increases beyond a safe limit, the follower accelerates to close the gap.

Operating Voltage	5V DC
Operating Current	15 mA
Frequency	40 kHz
Measuring Range	2 cm – 400 cm
Measuring Angle	15 degrees
Trigger Input Signal	10 μ s TTL pulse
Accuracy	± 3 mm
Pins	VCC, Trig, Echo, GND



[Figure 2.3 – HC-SR04 Ultrasonic Sensor]

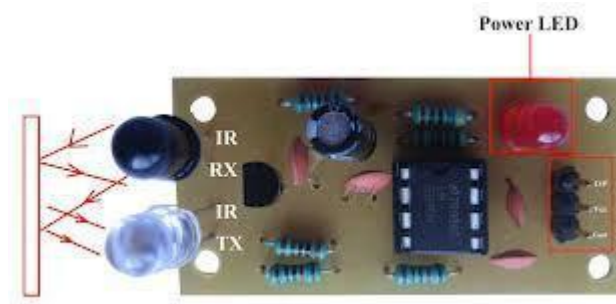
2.5 IR Sensor Modules (Left & Right)

Infrared (IR) sensors are used for detecting objects in the path of the vehicle and for directional line-following. The IR sensor module consists of an IR emitter LED and a photodiode receiver. When the emitter radiates infrared light and it reflects back from a nearby surface to the receiver, the output pin goes LOW (object detected). In open space, the output remains HIGH.

Role in the Truck Platooning System:

Two IR sensors are mounted on the front-left and front-right sides of the follower vehicle. They assist in detecting the edge of the path or lane boundaries, help the follower vehicle replicate the turning direction of the leader, and prevent the vehicle from going off course. In conjunction with the ultrasonic sensor, the IR sensors provide a comprehensive obstacle and direction detection system.

Operating Voltage	3.3V – 5V DC
Detection Range	2 cm – 30 cm (adjustable via potentiometer)
Output Type	Digital (HIGH/LOW)
LED Indicator	Power and detection indicator LEDs
Sensitivity	Adjustable via onboard trimmer
Interface	3-pin (VCC, GND, OUT)



[Figure 2.4 – IR Sensor Module]

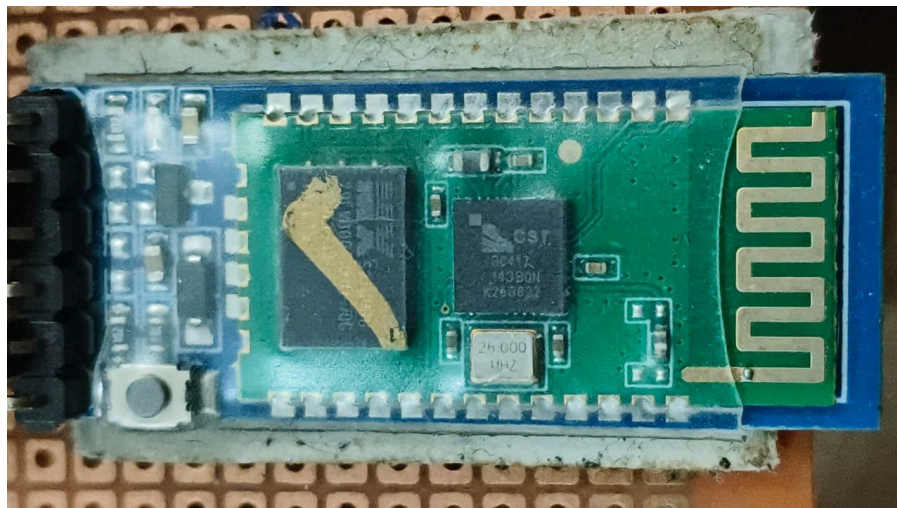
2.6 HC-05 Bluetooth Module

The HC-05 is a serial Bluetooth module that operates in the 2.4 GHz ISM band and supports Bluetooth v2.0+EDR (Enhanced Data Rate). It communicates with the Arduino Uno via UART serial communication at a default baud rate of 9600 bps. The HC-05 can be configured as either a Master or Slave device — in this project, the leader vehicle's HC-05 is configured as Master and the follower's as Slave, enabling automatic pairing and communication.

Role in the Truck Platooning System:

The leader truck transmits its movement commands (speed, direction) wirelessly to the follower truck via the HC-05 Bluetooth link. This communication enables the follower to replicate the leader's behavior in real time. The wireless range is typically 10 meters in open space, which is sufficient for prototype-scale platooning tests.

Bluetooth Version	v2.0+EDR
Operating Frequency	2.4 GHz ISM Band
Communication	UART (Serial)
Default Baud Rate	9600 bps
Operating Voltage	3.3V – 6V DC
Range	Up to 10 meters (Class 2)
Default PIN	1234
Modes	Master / Slave (configurable via AT commands)



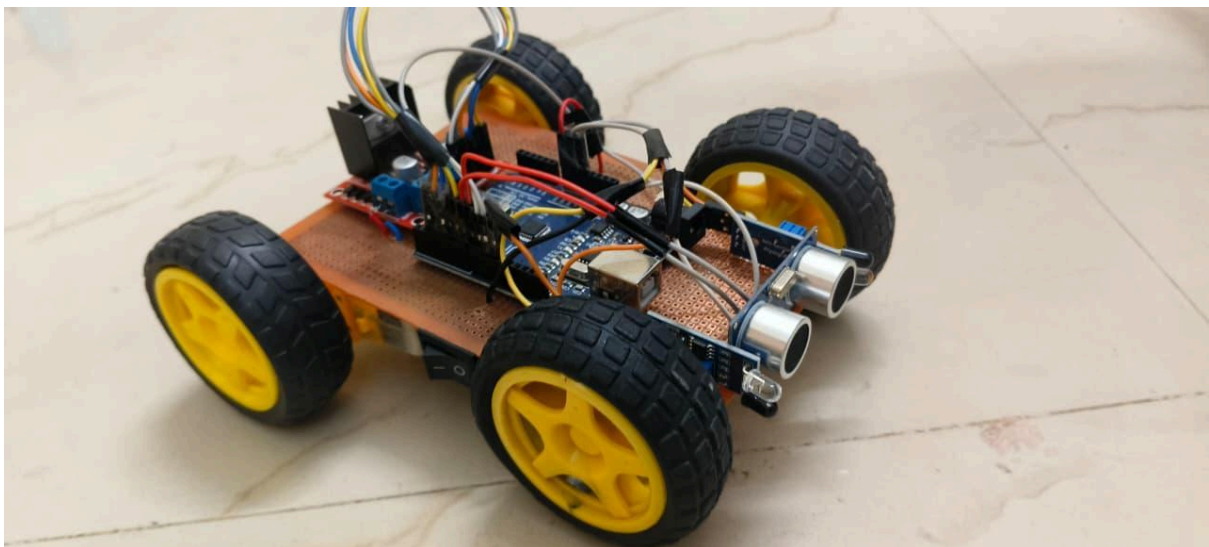
[Figure 2.5 – HC-05 Bluetooth Module]

2.7 DC Motors and Robot Chassis

Two TT Gear DC motors are used on each vehicle unit for locomotion. These are small, lightweight gearbox motors with yellow plastic bodies, commonly used in DIY robotics projects. They operate at 3–6V and provide sufficient torque for driving the robot chassis over flat surfaces.

The robot chassis is a 4-wheel acrylic or ABS plastic frame designed to accommodate two TT gear motors, caster wheels, battery holder, and electronics. It provides a sturdy mechanical base for assembling all components of each vehicle unit.

Motor Type	TT Gear DC Motor
Operating Voltage	3V – 6V DC
No-load Speed	~200 RPM at 5V
Stall Torque	~800 g·cm
Gear Ratio	1:48
Chassis Material	Acrylic / ABS Plastic
Wheel Diameter	65 mm
Chassis Size	~200mm × 150mm × 45mm



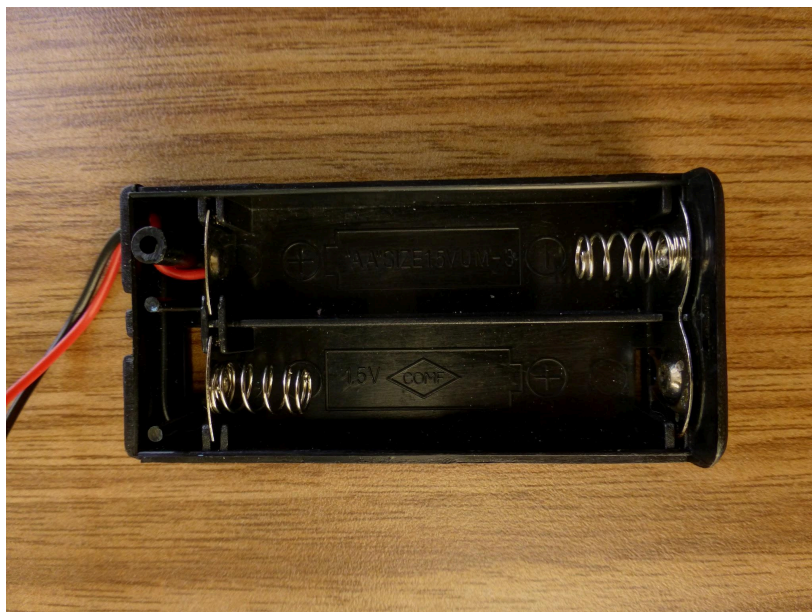
[Figure 2.6 – DC Motor and Robot Chassis]

2.8 Power Supply (Battery Pack)

Each vehicle unit requires a stable and sufficient power supply for the Arduino Uno, motor driver, motors, sensors, and Bluetooth module. A 7.4V Li-ion battery pack (2S configuration) or a 9V battery is used as the primary power source.

The battery connects to the L298N motor driver's VMS input, which has an onboard 5V voltage regulator to supply logic-level power to the Arduino Uno and sensors. Separate power distribution is ensured so that motor current surges do not affect the microcontroller operation.

Battery Type	Li-ion 7.4V (2S) or 9V Alkaline
Capacity	2000 mAh (Li-ion) typical
Output Voltage	7.4V – 9V nominal
Logic Supply	5V (via L298N onboard regulator)
Charging	USB or dedicated charger (Li-ion)
Protection	Over-charge and over-discharge (Li-ion BMS)

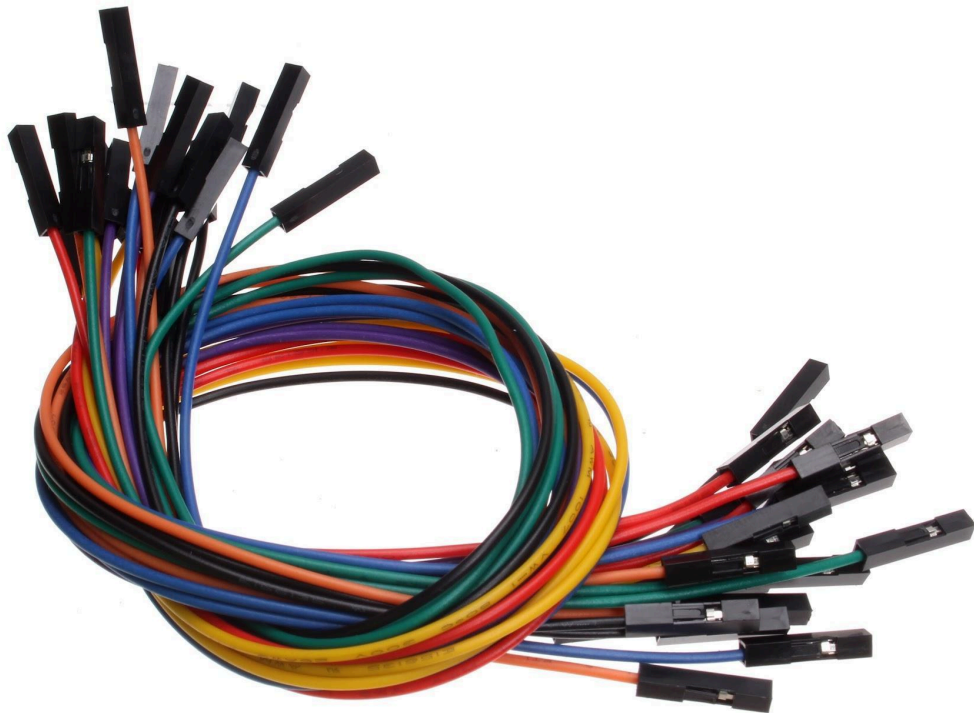


[Figure 2.7 – Battery Pack]

2.9 Jumper Wires and Toggle Switch

Jumper wires (male-to-male, male-to-female, and female-to-female) are used to make all electrical connections between the Arduino Uno, sensors, motor driver, and Bluetooth module. They allow flexible and easy re-routing of connections during prototyping and testing without the need for soldering.

A toggle switch or slide switch is incorporated in the power circuit of each vehicle to allow convenient power ON/OFF control without disconnecting the battery. This makes it practical during testing and demonstrations.



CHAPTER 3: ARDUINO SOFTWARE

3.1 Introduction to Arduino IDE

The Arduino Integrated Development Environment (IDE) is the official software tool used to write, compile, and upload programs to Arduino boards. It provides a simple and accessible platform for developing embedded system applications, making it ideal for this project.

The Arduino IDE supports a simplified version of C/C++ programming language augmented with a rich library ecosystem. For this project, the IDE is used to write programs for both the leader and follower Arduino Uno boards. The software tools used include:

- Arduino IDE (version 1.8.x or 2.x) for code development and uploading
- Built-in Serial Monitor for real-time debugging and Bluetooth data visualization
- NewPing library (optional) for efficient HC-SR04 ultrasonic sensor interfacing
- SoftwareSerial library for additional UART communication channels

3.2 Programming Logic – Leader Vehicle

The leader vehicle's Arduino Uno is programmed to:

1. Receive movement commands from a user via the HC-05 Bluetooth module (from a mobile phone app or another Arduino).
2. Map the received commands to specific motor control instructions (forward, backward, left, right, stop).
3. Transmit the current movement state to the follower vehicle via the secondary HC-05 module.
4. Drive the L298N motor driver to move the leader chassis accordingly.

3.3 Programming Logic – Follower Vehicle

The follower vehicle's Arduino Uno is programmed to:

5. Continuously measure the distance to the vehicle ahead using the HC-SR04 sensor.
6. Read the direction commands received from the leader via Bluetooth communication.
7. Read the left and right IR sensor outputs to detect lane or obstacle conditions.
8. Apply a distance-based control algorithm to maintain the safe following gap.
9. Output appropriate PWM signals to the L298N driver for speed control and direction.
10. Trigger an emergency stop if an obstacle is detected within a critical distance threshold.

3.4 Motor Control Strategy

Speed control of the DC motors is achieved through PWM signals sent to the Enable pins (ENA, ENB) of the L298N module. The Arduino analogWrite() function generates PWM signals with duty cycles ranging from 0 (stopped) to 255 (full speed). Direction control is managed by setting the IN1-IN4 digital pins HIGH or LOW in the appropriate combinations:

Direction	IN1	IN2	IN3	IN4	ENA / ENB
Forward	HIGH	LOW	HIGH	LOW	PWM (0–255)
Backward	LOW	HIGH	LOW	HIGH	PWM (0–255)
Turn Left	LOW	HIGH	HIGH	LOW	PWM (0–255)
Turn Right	HIGH	LOW	LOW	HIGH	PWM (0–255)
Stop	LOW	LOW	LOW	LOW	0

CHAPTER 4: PROPOSED MODEL

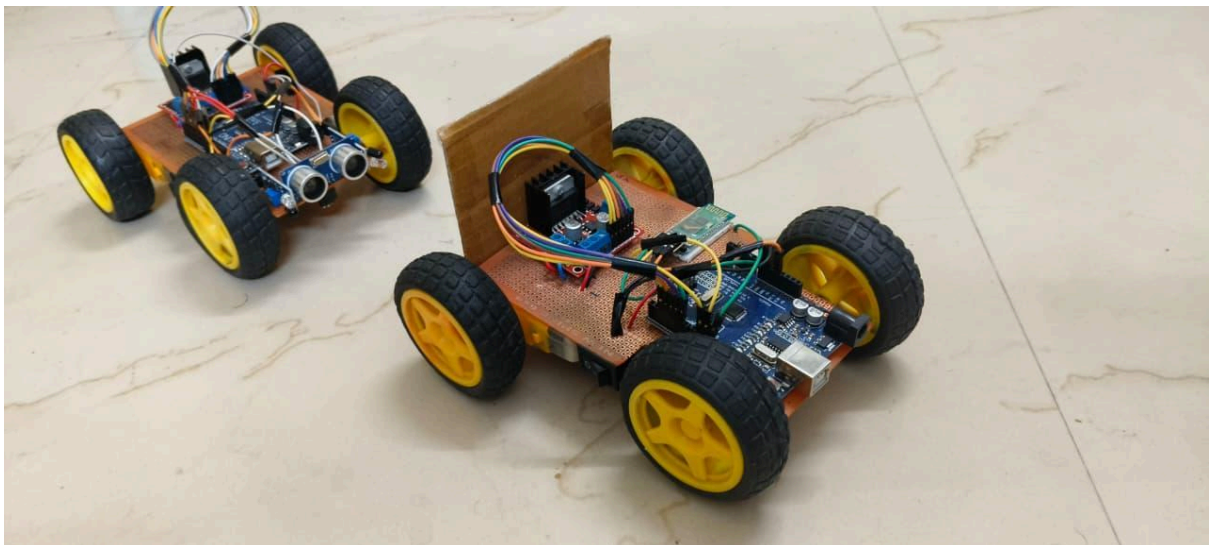
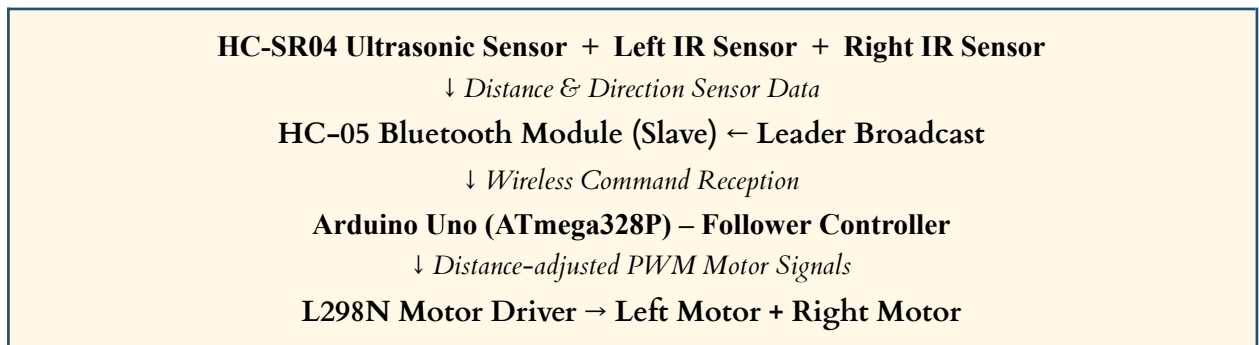
4.1 Block Diagram

The Truck Platooning System consists of two vehicle units — the Leader Truck and the Follower Truck — each controlled by an Arduino Uno. The system architecture is described below:

LEADER TRUCK:



FOLLOWER TRUCK:



[Figure 4.1 – System Block Diagram of Truck Platooning System]

4.2 Circuit Connections – Pin Diagram

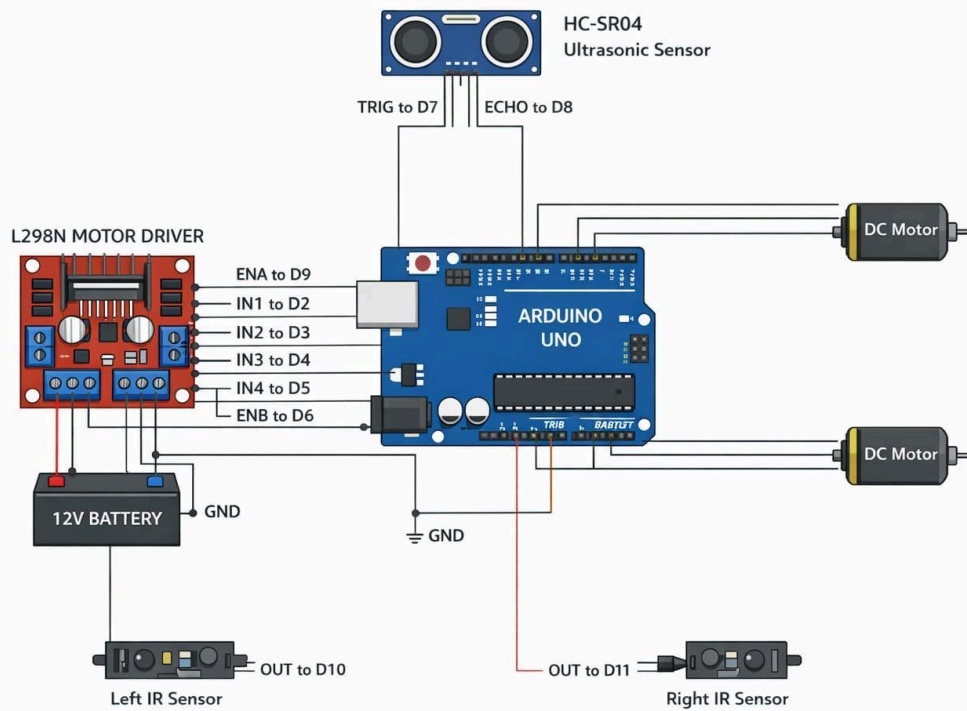
Leader Vehicle – Arduino Uno Pin Connections:

Component	Arduino Pin	Notes
L298N – IN1	D2	Left motor direction A
L298N – IN2	D3	Left motor direction B
L298N – IN3	D4	Right motor direction A
L298N – IN4	D5	Right motor direction B
L298N – ENA	D6 (PWM)	Left motor speed control
L298N – ENB	D9 (PWM)	Right motor speed control
HC-05 – TX	D10 (RX)	Bluetooth RX (via SoftwareSerial)
HC-05 – RX	D11 (TX)	Bluetooth TX (via SoftwareSerial) – Use 1k/2k divider
HC-05 – VCC	5V	Power supply for HC-05
HC-05 – GND	GND	Common ground
L298N – VMS	Battery (+)	Motor power supply (7.4–9V)
L298N – GND	Battery (-) + Arduino GND	Common ground reference

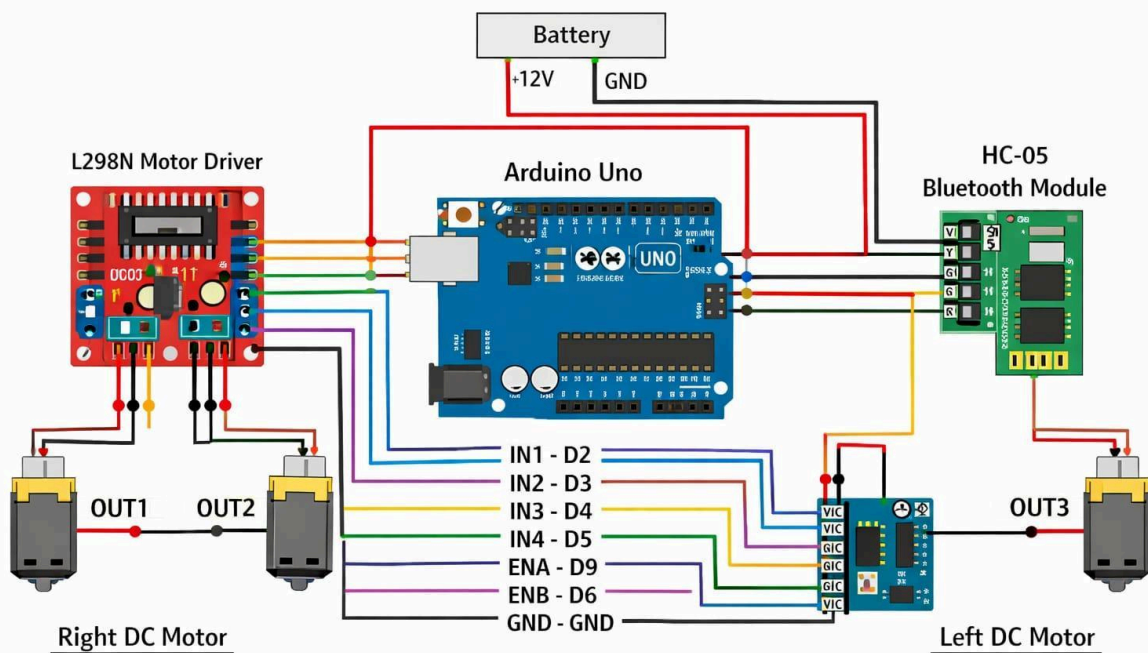
Follower Vehicle – Additional Sensor Connections:

Component	Arduino Pin	Notes
HC-SR04 – Trig	D7	Trigger pulse output
HC-SR04 – Echo	D8	Echo pulse input
HC-SR04 – VCC	5V	Sensor power
HC-SR04 – GND	GND	Common ground
Left IR Sensor – OUT	D12	Left obstacle detection input
Right IR Sensor – OUT	D13	Right obstacle detection input
IR Sensors – VCC	5V	Sensor power
IR Sensors – GND	GND	Common ground

Follower Robot Circuit Diagram



Leader Truck Circuit



[Figure 4.2 – Circuit Diagram of Truck Platooning System]

Note: In the actual circuit diagram, the follower vehicle also includes all connections from the leader vehicle table above, plus the additional sensor connections listed here. A voltage divider ($1k\Omega$ and $2k\Omega$ resistors) must be used between Arduino TX and HC-05 RX to protect the Bluetooth module from 5V signals.

4.3 Working Principle / Algorithm

System Startup:

11. Both leader and follower vehicle Arduino boards are powered ON.
12. HC-05 Bluetooth modules pair automatically (Master-Slave configuration).
13. All servo/motor positions initialize; sensors begin continuous reading.

Leader Truck Operation:

14. The user sends the direction command (F/B/L/R/S) via mobile Bluetooth app.
15. HC-05 receives command and forwards to Arduino via UART.
16. Arduino maps command to motor control signals.
17. L298N drives motors in the corresponding direction.
18. Arduino simultaneously broadcasts the command to the follower truck's HC-05.

Follower Truck Operation:

19. HC-05 receives a broadcast command from the leader.
20. HC-SR04 measures the current distance to the vehicle ahead.
21. Arduino checks: if distance < MIN_SAFE_DIST → STOP/SLOW DOWN.
22. Arduino checks: if distance > MAX_SAFE_DIST → SPEED UP.
23. IR sensors checked: if left sensor triggered → steer right; if right → steer left.
24. L298N drives motors according to computed speed and direction.
25. Loop continues at ~50ms intervals for smooth real-time response.

CHAPTER 5: CONNECTIONS AND RESULTS

5.1 Hardware Connections

The hardware assembly for each vehicle unit is carried out as follows. The Arduino Uno is securely mounted on the robot chassis as the central controller. The L298N motor driver module is connected to both DC motors and the battery power supply. Motor control signals from the Arduino's digital and PWM pins (D2–D6, D9) are fed into the L298N IN1–IN4 and ENA/ENB pins respectively.

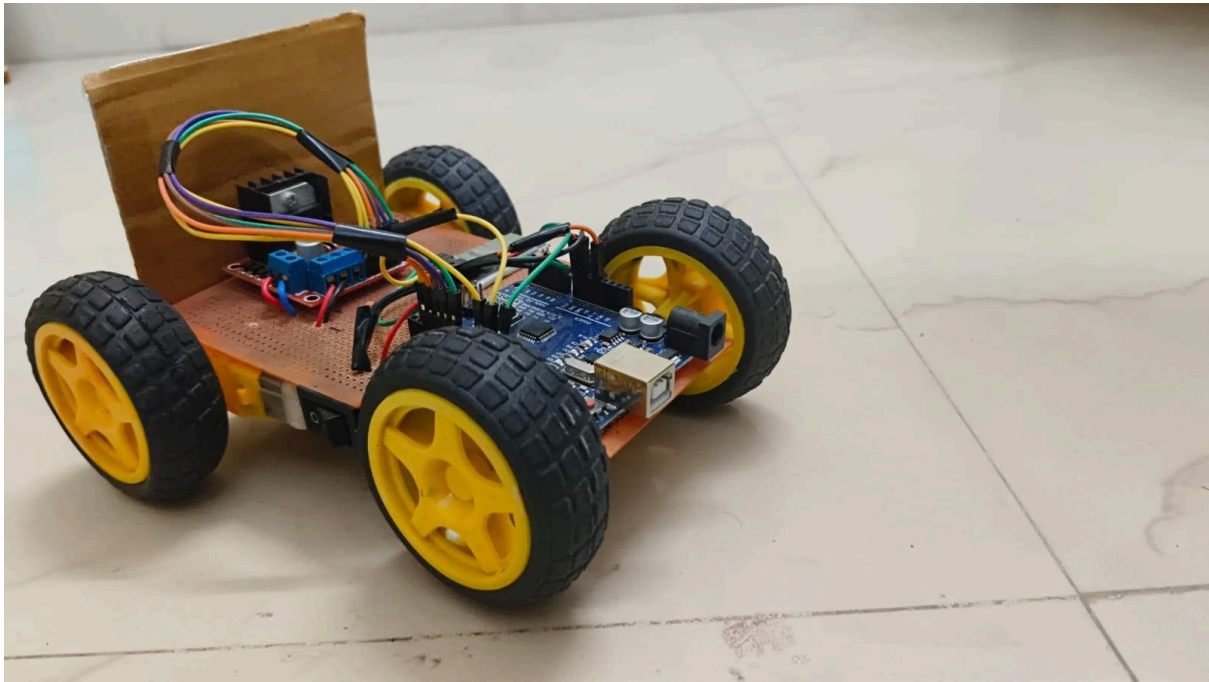
The HC-05 Bluetooth module is interfaced to the Arduino via SoftwareSerial on pins D10 (RX) and D11 (TX). A resistor voltage divider (1k Ω and 2k Ω) is placed between Arduino TX (D11) and HC-05 RX to step down the 5V signal to approximately 3.3V, protecting the Bluetooth module. The HC-05 is powered from the Arduino's 5V pin.

On the follower vehicle, the HC-SR04 ultrasonic sensor is mounted at the front of the chassis. Its Trig and Echo pins are connected to Arduino digital pins D7 and D8 respectively. The two IR sensor modules are mounted on the front-left and front-right of the chassis, with their output signals connected to D12 and D13.

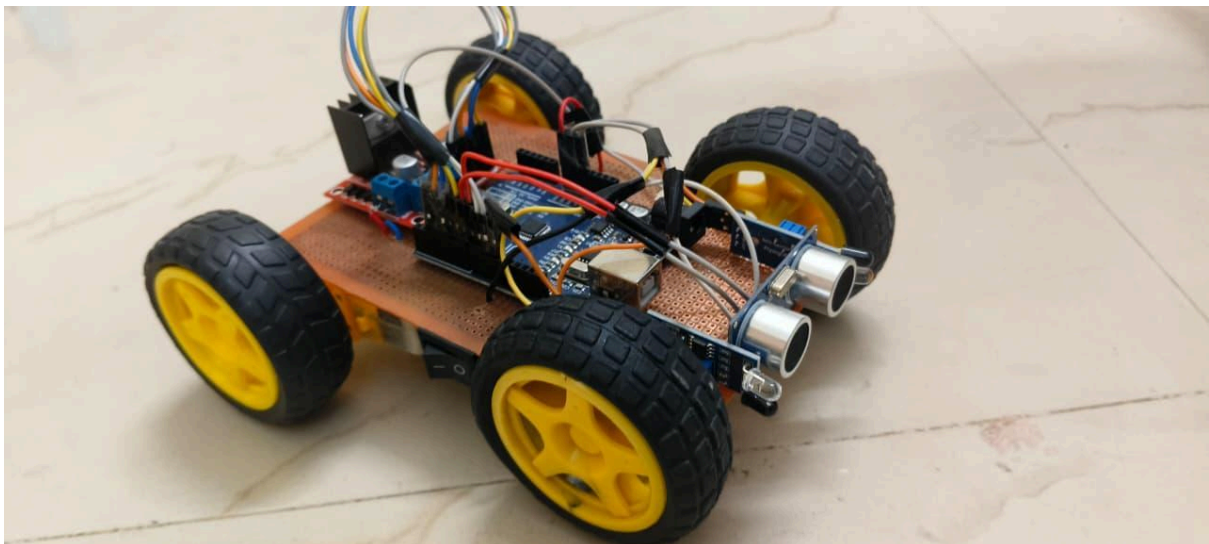
All components share a common ground. The battery (7.4V Li-ion or 9V) connects to the L298N's motor power input (VMS), which steps down 5V for logic-level components via its onboard regulator. The toggle switch is placed in series with the battery's positive terminal for convenient power control.

Important Wiring Notes

1. Always connect all GND rails together (Arduino GND, L298N GND, Sensor GND).
2. Use a $1\text{k}\Omega$ - $2\text{k}\Omega$ resistor voltage divider between Arduino TX and HC-05 RX.
3. Do not power high-current motors directly from Arduino 5V pin — use the L298N.
4. Ensure the battery is sufficiently charged before testing for accurate sensor readings.



[Figure 5.1 – Hardware Prototype: Leader Truck Assembly]



[Figure 5.2 – Hardware Prototype: Follower Truck Assembly]

5.2 Results

The Truck Platooning System prototype was successfully assembled and tested under controlled laboratory conditions. The following results were observed during testing:

Bluetooth Communication:

The HC-05 modules successfully paired in Master-Slave mode. Commands transmitted from the mobile Bluetooth application to the leader truck were reliably received and processed. The wireless range of approximately 8–10 meters was achieved in an open indoor environment.

Distance-Based Speed Control:

The follower truck accurately maintained a safe following distance using the HC-SR04 ultrasonic sensor. When the leader truck moved forward, the follower automatically accelerated to close the gap. When the leader stopped, the follower decelerated and halted within the safe distance threshold (approximately 15–20 cm in the prototype scale).

Direction Tracking:

The follower truck successfully replicated left and right turn commands from the leader. The IR sensors provided supplementary obstacle detection and prevented the follower from deviating from its path.

Emergency Stop:

When an obstacle was placed between the two vehicles within the critical distance (< 10 cm in prototype scale), the follower truck immediately stopped, demonstrating the emergency stop safety mechanism.

Test Parameter	Observed Value	Status
Bluetooth pairing range	~10 meters	✓ Pass
Command response latency	< 200 ms	✓ Pass
Safe following distance maintained	15–20 cm (prototype)	✓ Pass
Emergency stop distance	< 10 cm	✓ Pass
Direction replication accuracy	$> 90\%$	✓ Pass
Continuous operation time	~45 minutes (7.4V 2000mAh)	✓ Pass

CHAPTER 6: APPLICATIONS AND CONCLUSION

6.1 Applications of Truck Platooning System

- Long-distance highway freight transport with reduced driver requirements.
- Mining and quarry site vehicle convoy systems for transporting extracted materials.
- Industrial warehouse logistics for autonomous internal goods transportation.
- Agricultural applications such as automated harvesting convoys.
- Military logistics for unmanned supply chain convoy operations.
- Port and terminal operations for container movement between loading areas.
- Construction site material transportation between storage and work zones.
- Smart city delivery systems for last-mile logistics automation.
- Emergency response convoy systems for disaster relief operations.
- Educational robotics platforms for demonstrating autonomous vehicle principles.
- Research prototype for developing full-scale autonomous truck platooning systems.

6.2 Conclusion

The Truck Platooning System using Arduino Uno has been successfully designed, implemented, and tested as a prototype. The system effectively demonstrates the core principles of vehicle platooning — autonomous following, inter-vehicle distance maintenance, wireless communication-based coordination, and safety through obstacle detection.

The integration of the HC-SR04 ultrasonic sensor, IR sensors, HC-05 Bluetooth module, and L298N motor driver with Arduino Uno provides a reliable, low-cost, and educationally valuable platform for exploring autonomous vehicle concepts.

The system successfully achieved all stated objectives: the follower truck autonomously tracked the leader, maintained safe distance, replicated directional commands, and implemented emergency stopping. The Bluetooth communication provided a stable and responsive wireless link between the two vehicles.

This project demonstrates that even with affordable off-the-shelf components, meaningful autonomous vehicle behavior can be implemented, providing a strong foundation for developing more advanced full-scale truck platooning solutions.

6.3 Future Scope

Hardware Improvements:

- Upgrade to more powerful motors and a sturdier chassis to carry heavier payloads.
- Add encoders to DC motors for precise speed measurement and closed-loop control.
- Integrate GPS modules for outdoor tracking and absolute position awareness.

Communication Improvements:

- Replace Bluetooth with Wi-Fi (ESP8266/ESP32) or LoRa modules for longer range.
- Implement V2V (Vehicle-to-Vehicle) communication protocol for multi-truck platoons beyond 2 vehicles.
- Add a central server or cloud dashboard for real-time fleet monitoring.

Intelligence and Autonomy:

- Integrate a camera module (e.g., OV7670 or Raspberry Pi Camera) for computer vision-based tracking.
- Implement PID (Proportional-Integral-Derivative) control for smoother speed regulation.
- Apply machine learning algorithms for predictive behavior and adaptive platooning.
- Add LiDAR sensor for 360-degree obstacle mapping and advanced collision avoidance.
- Scale up to 3+ vehicle platoons with hierarchical control architecture.

CHAPTER 7: REFERENCES

26. Banzi, M., & Shiloh, M., Getting Started with Arduino, Maker Media Publications.
27. John-David Warren, Josh Adams, and Harald Molle, Arduino Robotics, Apress Publications.
28. Simon Monk, Programming Arduino: Getting Started with Sketches, McGraw Hill Education.
29. Arduino Official Website – <https://www.arduino.cc>
30. HC-SR04 Ultrasonic Sensor Datasheet – Cytron Technologies.
31. L298N Motor Driver Module Datasheet – STMicroelectronics.
32. HC-05 Bluetooth Module AT Commands and Datasheet.
33. ATmega328P Microcontroller Datasheet – Microchip Technology.
34. Electronics Tutorials Website – <https://www.electronics-tutorials.ws>
35. Bergenhem, C. et al., Overview of Platooning Systems, 19th ITS World Congress, Vienna, 2012.
36. Tsugawa, S., Kato, S., Aoki, K., An Automated Truck Platoon for Energy Saving, 2011 IEEE/RSJ International Conference on Intelligent Robots and Systems.
37. Various online tutorials and educational resources related to Arduino Uno, L298N Motor Driver, HC-SR04 sensor, and Robotics Projects.

CHAPTER 8: APPENDIX – ARDUINO PROGRAM

Leader Vehicle – Arduino Code

```
#include <SoftwareSerial.h>

SoftwareSerial bt(10,11);

int IN1 = 2; // Right motor
int IN2 = 3;
int IN3 = 4; // Left motor
int IN4 = 5;
int ENA = 9;
int ENB = 6;
char command;
int speedVal = 170;

void setup() {

  bt.begin(9600);
  Serial.begin(9600);

  pinMode(IN1,OUTPUT);
  pinMode(IN2,OUTPUT);
  pinMode(IN3,OUTPUT);
  pinMode(IN4,OUTPUT);

  pinMode(ENA,OUTPUT);
  pinMode(ENB,OUTPUT);
}

void loop() {

  if(bt.available()){

    command = bt.read();
    Serial.println(command);

    switch(command){

      case 'F': forward(); break;
      case 'B': backward(); break;
      case 'L': left(); break;
      case 'R': right(); break;
      case 'S': stopCar(); break;

    }
  }
}
```



```

void forward(){

    analogWrite(ENA, speedVal);
    analogWrite(ENB, speedVal);

    digitalWrite(IN1,HIGH);
    digitalWrite(IN2,LOW);
    digitalWrite(IN3,HIGH);
    digitalWrite(IN4,LOW);
}

void backward(){

    analogWrite(ENA, speedVal);
    analogWrite(ENB, speedVal);

    digitalWrite(IN1,LOW);
    digitalWrite(IN2,HIGH);
    digitalWrite(IN3,LOW);
    digitalWrite(IN4,HIGH);
}

void left(){ // RIGHT motor runs

    analogWrite(ENA,speedVal);
    analogWrite(ENB,0);

    digitalWrite(IN1,HIGH);
    digitalWrite(IN2,LOW);
}

void right(){ // LEFT motor runs

    analogWrite(ENA,0);
    analogWrite(ENB,speedVal);

    digitalWrite(IN3,HIGH);
    digitalWrite(IN4,LOW);
}

void stopCar(){

    analogWrite(ENA,0);
    analogWrite(ENB,0);

}

```

Follower Vehicle – Arduino Code

```
// MOTOR PINS
int IN1 = 2;
int IN2 = 3;
int IN3 = 4;
int IN4 = 5;
int ENA = 9;
int ENB = 6;
int trig = 7;// ULTRASONIC
int echo = 8;
int leftIR = 10;// IR SENSORS
int rightIR = 11;
long duration;
int distance;

void setup() {

Serial.begin(9600);

pinMode(IN1,OUTPUT);
pinMode(IN2,OUTPUT);
pinMode(IN3,OUTPUT);
pinMode(IN4,OUTPUT);

pinMode(ENA,OUTPUT);
pinMode(ENB,OUTPUT);

pinMode(trig,OUTPUT);
pinMode(echo,INPUT);

pinMode(leftIR,INPUT);
pinMode(rightIR,INPUT);

analogWrite(ENA,200);
analogWrite(ENB,200);

}

void loop() {

// Measure distance
```

```

digitalWrite(trig,LOW);
delayMicroseconds(2);

digitalWrite(trig,HIGH);
delayMicroseconds(10);
digitalWrite(trig,LOW);

duration = pulseIn(echo,HIGH);
distance = duration * 0.034 / 2;

int left = digitalRead(leftIR);
int right = digitalRead(rightIR);

// DISTANCE CONTROL
if(distance < 10){
    backward();
}

else if(distance > 20){

    if(left == 0 && right == 1)
    {
        turnLeft();
    }

    else if(left == 1 && right == 0)
    {
        turnRight();
    }

    else{
        forward();
    }

}

else{
    stopCar();
}

delay(40);
}

```

```
void forward(){

digitalWrite(IN1,LOW);
digitalWrite(IN2,HIGH);
digitalWrite(IN3,LOW);
digitalWrite(IN4,HIGH);

}

void backward(){

digitalWrite(IN1,HIGH);
digitalWrite(IN2,LOW);
digitalWrite(IN3,HIGH);
digitalWrite(IN4,LOW);

}

void turnLeft(){

digitalWrite(IN1,LOW);
digitalWrite(IN2,HIGH);
digitalWrite(IN3,LOW);
digitalWrite(IN4,LOW);

}

void turnRight(){

digitalWrite(IN1,LOW);
digitalWrite(IN2,LOW);
digitalWrite(IN3,LOW);
digitalWrite(IN4,HIGH);

}

void stopCar(){

digitalWrite(IN1,LOW);
```

```
digitalWrite(IN2,LOW);  
digitalWrite(IN3,LOW);  
digitalWrite(IN4,LOW);  
}
```

Program Explanation

The Leader Vehicle code sets up the L298N motor driver and HC-05 Bluetooth module. In the main loop, it continuously checks for incoming Bluetooth commands (F = Forward, B = Backward, L = Left, R = Right, S = Stop) from a mobile application. The executeCommand() function maps each character to the appropriate motor control routine (moveForward, moveBackward, turnLeft, turnRight, stopMotors). Each command is also broadcast to the follower vehicle via the same Bluetooth serial link.

The Follower Vehicle code integrates three types of inputs: Bluetooth commands from the leader, ultrasonic distance measurements from the HC-SR04 sensor, and obstacle detection from left and right IR sensors. The control logic implements a priority-based decision system. Emergency stop is triggered first when distance falls below 8 cm. For forward commands, speed is dynamically adjusted based on inter-vehicle distance — slowing down if too close, speeding up if too far. Direction commands from the leader are replicated, with IR sensor data providing supplementary steering correction. A 50ms control loop delay ensures stable and smooth motor response.

Safety Note

The constrain() function can be used to limit speed values within safe operating ranges (0–255 for PWM). Always test the system in an open area away from people when first powered on. Ensure all connections are secure and no bare wires are exposed before powering the circuit.